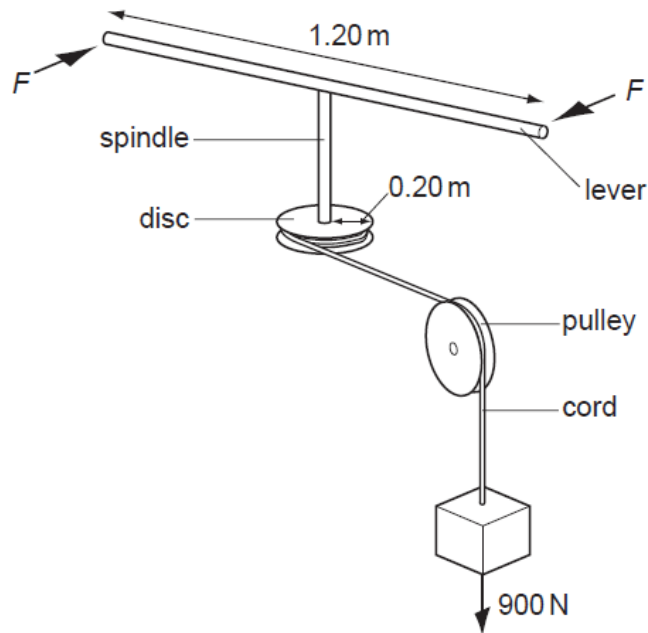


- 6** One end of a spindle is attached to the centre of a lever of length 1.20 m and its other end is attached to the centre of a disc of radius 0.20 m as shown in the figure below.

A cord is wrapped around the disc, passes over a pulley and is attached to a 900 N weight at one end.



The mass of the lever, spindle, disc, cord and pulley is assumed to be negligible. Equal and opposite forces of magnitude F is applied to each end of the lever.

Ignoring frictional forces, what is the minimum value of F needed to balance the 900 N weight?

- | | |
|----------|-------|
| A | 75 N |
| B | 150 N |
| C | 300 N |
| D | 950 N |

Ans: B

Tension in the cord = 900 N
 Torque on disc = $900 \times 0.20 = 180$ Nm
 Torque due to $F = F \times 1.20 = 180$
 $F = 180/1.2 = 150$ N